

Bayero University Kano

Kano

Post UTME Past Questions and Answers

MATHEMATICS



Compiled

by

CLASSIC EDUCATIONAL CONSULTS

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ABOUT CLASSIC EDUCATIONAL CONSULTS

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TIPS TO PASS POST UTME EXAMS

Do you want to pass post UTME of any tertiary institution in Nigeria? If yes, then you are in the right place. In this write up I will teach you best tips on how to prepare for Post UTME of any university or polytechnic in Nigeria. I enjoin you to read thoroughly, as I will highlight the ultimate preparation tips to pass post UTME screening examinations.

- **Have the mindset of a Success:**

Why is this, the first step in today's guideline? Having the mindset of a Successful student is first and surest step towards any form of progress in life. Be motivated to pass your Post UTME this year. If you don't continually prepare yourself and mind to pass this exam, there is nothing someone else can do to prepare you. Don't take what comes, go for what you want. Desire a Post UTME score and work towards making it a reality. Always tell yourself, "I must pass this Post UTME examination once and for all".

- **Don't get discouraged by your JAMB Score:**

Many a times, one of the main reasons why a lot of candidates fail to make it through the Post UTME exams is as a result of low self esteem due to their JAMB scores. One thing is to Score high marks in your JAMB Examination, another is to score high and beat the School's Departmental cut off marks. However, you need to develop a positive attitude toward your Post UTME irrespective of your Jamb score. Always strive to pass Post UTME examination irrespective of what you think you scored in Jamb. Post UTME is a very important factor that determines who gains admission. Even if you scored low in Jamb, you still have a golden opportunity to study your desired course at the University. A very high Post UTME score can help boost your Jamb result. One of the greatest mistakes candidates make in post utme is thinking that their high Jamb score is enough. Don't be deceived, no matter how high your Jamb score may be, a low Post UTME score will disgrace it.

- **Gather Relevant Information:**

A lot of candidates have insufficient or poor information about how the exam will play out, they have no clue whatsoever of how many questions they will be asked, what form the examination will be set in; and it is for these reasons that having relevant information is paramount. You should know beforehand whether you are writing a paper based test or a computer based examination, your information gathering will help you know how much time is allotted to your paper, and will help you while studying to time yourself as you practice questions, giving you an edge over thousands of candidates who only find out about the time allotted for the paper only minutes before the exam commences. Also, you should check to know the terrain of the school. If possible, travel to the school and ask around. Ensure you have somewhere to stay a day or two before your examination (if out of state). This is important since you need to be well-rested before sitting for an examination if you hope to give it your best and perform well. Finally, try to gather as much information as you can about your university and department of choice, especially such things as required subject combination and cut-off mark from the previous year; as this could prove the difference between you and other inexperienced candidates, and whether you get an admission to your preferred course of study or not.

- **Avoid Procrastination:**

Later things... This is the language of procrastinators. They always think there is time. My dear, do whatever you need to do now. Time is an asset and not a handset. The later you always say doesn't come as expected. Learn to live your life with a sense of urgency. Have the mentality, "I am behind schedule...". Your Post UTME examination is not far. Say no to procrastination.

- **Study with Past-Questions:**

Dear friend, do not fall into the misguided conception that you can go it alone! It is important to have a clear understanding of the questions you are likely to be asked and the likely areas the questions might come from, and what better way to do that than with questions of previous examinations? How do I get Post UTME past questions for my University of choice? It is very easy. Yes, it is! We have compiled FREE and RELIABLE post UTME past questions and answers for various Institutions. So, do try your best to get relevant past-questions that apply to your school and department, and go through it each day.

- **Set your Target:**

Your background check should tell you what the cut-off mark of your department of choice, hence your target point. You have to set a target for yourself and determine in your heart that you can achieve it, as you work towards it. Make sure the only thing that robs you of sleep is how to successfully meet your target score. Ensure that whatever target you set is above what you scored in UTME, since you will need to do better than your UTME performance to make the merit list of your department.

- **Reduce your time in Social Media:**

Some people live their entire life on Facebook, Twitter, Instagram, Whatsapp etc. This is so bad and disastrous if you are preparing for exams. Try and reduce you time spending on social media during this time. Maybe, after the exams you can go back and make it your second home, (just saying).

- **Learn To Remember:**

To pass Post UTME once and for all, you must learn to remember. It is not just enough to read. It is more important to remember what you have read. The difference between someone who reads for shorter hours but pass very well and another who studies for long hours but don't perform well is "remembrance". Study to remember and not to forget. If you can remember all you have read, then success will smile on you. Do whatever it takes to pass Post UTME examination genuinely.

- **Learn To Solve Without a Calculator:**

I know you cannot give me the correct answer to $\frac{2}{3}$ right now without a calculator. I can see that you are trying to get the answer. No, don't give me that look, I know just what I'm saying. Most schools do not allow the use of calculator in their Post UTME examination. Therefore, you must learn to solve problems without pressing your calculator. It is time to temporarily break the covenant you have with calculator.

- **Give yourself enough time to rest:**

Another factor that causes low exam performance is STRESS! Stress as we all know it is very disastrous to the Human body. You need to give your body the maximum amount of rest it needs. Reading, reading and reading all the time isn't advisable at all, according to science, the Human brain is always at work. Give your brain time to relax from the previous tasks so as to recover and work efficiently.

NOTE: We can't cover everything in this article; we'll be continuing this discussion some other day.

Don't forget to share your thoughts with us @ classicconsults2005@gmail.com

Classic Consults wishes you good Success in your examination

GENERAL NOTE

Benefits of Studying Past Exam Papers

Studying past papers is a valuable part of exam preparation and help keep revision **focused** on important themes whilst practicing exam style questions.

Past exam papers are one of the most helpful tools available to prepare for both internal and external examinations as they provides students with **practical insight** into how the forthcoming exam paper is likely to look and the **key themes** or **subject areas** most likely to be covered. Before starting revision for a subject it is always worth taking time to seek out past papers.

Please note that this is a post UTME past question and not post UTME Sure Questions and therefore it is meant to be practice and also guide you in your revision. The number of questions here may not be the exact number of questions you will be given in your case. Also the time you will be given may vary from the one indicated in this text. Therefore, Classic Educational Consults will not be responsible for any form of Examination Malpractice or Examination Misconduct.

POST UTME

1. What is the probability of having an odd number in a single toss of a fair die?
(a) $1/6$ (b) $1/3$ (c) $1/2$ (d) $2/3$ (e) $5/6$
2. A sales girl gave a change of N 1.15 to a customer instead of N 1.25. Calculate the Percentage error. (a) 10% (b) 8.7% (c) 8.0% (d) 2.4% (e) 0.10%
- 3.

Score	0	1	2	3	4	5
Frequency	2	3	4	2	7	2

Find the mean score in the frequency table above showing the performance a group of student in a test. (a) 1.75 (b) 2 (c) 2.5 (d) 2.75 (e) 3

4. Simplify $0.63954/0.003$ giving your answer correct to two significant figures.
(a) 213.18 (b) 213.00 (c) 213 (d) 210 (e) 2
5. What is the difference in longitude between P (lat. 50°N , long. 50°W) and Q (lat. 50°N , long. 150°W) (a) 300° (b) 200° (c) 130° (d) 100° (e) 300°

6. Simplify $(27^{1/3})^2$. (a) $4\frac{1}{2}$ (b) 6 (c) 9 (d) 18 (e) 81
7. If $\log_{10} a = 4$, what is a? (a) 0.4 (b) 40 (c) 400 (d) 1000 (e) 10000
8. Simplify $36^{1/2} \times 64^{-1/2} \times 5^\circ$ (a) 0 (b) $1/24$ (c) $2/3$ (d) $11/2$ (e) $7\frac{1}{2}$
9. Solve the equation $3a + 10 = a^2$ (a) $a = 5$ or $a = 2$ (b) $a = -5$ or $a = 2$ (c) $a = 10$ or $a = 0$
(d) $a = 5$ or $a = -2$ (e) $a = -5$

10. Solve $6(x-4) + 3(x+7) = 3$ (a) $3/2$ (b) $2/3$ (c) $1/2$ (d) $1/3$ (e) 0

11. The common ratio of a GP is 2. If the 5th term is greater than the 1st term by 45 find the 5th term. (a) 3 (b) 6 (c) 45 (d) 48 (e) 90

12. A house bought for N100, 000 was later auctioned for N 80,000. Find the loss percent. (A) 20% (b) 30% (c) 40% (d) 50% (e) 60%

13. Simplify $2 \log_3 6 + \log_3 12 - \log_3 16$ (A) 2 (b) 3 (c) $22 \log_3 2$ (d) $3 \log_3 2$ (e) $4 \log_3 2$

14. If $8x - 4 = 6x - 10$, find the value of $5x$. (a) -35 (b) -15 (c) -3 (d) 3 (e) 7

15. Given that $2p - m = 6$ and $2p + 4m = 1$. Find the value of $(4p + 3m)$. (a) 1 (b) 3 (c) 5 (e) 9

16. Which of the following is not a factor of $2p^2 - 2$. (a) 2 (b) $p-1$ (c) $p+1$ (d) $2p-2$ (e) $2p+1$

17. A car is traveling at an average speed of 80 km/hr. Its speed in meters per Second (m/s) is (a) 22.2m/s (b) 133.3m/s (c) 13.3m/s (d) 222.2m/s

18. Solve the equation $\log_8 x + 4 \log_8 x = 2$ (a) $1/4$ (b) $1/2$ (c) 1 (d) 2 (e) 4
19. Find the product of x and y if $x = 2/3$, $6/7$, y are in GP (a) $24/49$ (b) $4/7$ (c) $2/7$ (d) $7/4$ (e) $2 \frac{1}{8}$
20. A box contains 2 white and 3 blue identical balls. If two balls are picked at random one after the other without replacement, what is the probability of picking two balls of different colours? (a) $6/25$ (b) $7/20$ (c) $3/5$ (d) $2/3$ (e) $5/6$
21. For what value of y is the expression $y + 2$ undefined? $Y^2 - 3y - 10$ (a) $Y = 0$ (b) $y = 2$ (c) $y = 3$ (d) $y = 5$ (e) $y = 10$
22. A piece of cloth was measured as 6.10m. If the actual length of the cloth is 6.35m, find the percentage error correct to 2 decimal places. (a) 3.05% (b) 3.94% (c) 15.00% (d) 25.00% (e) 39.30%
23. The radius of a geographical globe is 60cm. Find the length of the parallel of latitude $60^\circ N$ (a) 66cm (b) 60cm (c) 30cm (d) 10cm (e) 6cm
24. A die with faces numbered 1 to 6 is rolled once. What is the probability of obtaining 1 or 4? (a) $1/36$ (b) $1/6$ (c) $1/3$ (d) $1/2$ (e) $2/3$
25. Make S the subject of the formula $V = K/\sqrt{T-S}$ (a) $S = T - (K^2/V^2)$ (b) $S(K^2/V^2) = T$
 (c) $S = T - (K^2/V^2)$ (d) $S = T(V^2 - K^2)/K^2$ (e) $S = T + (V^2/K^2)$

POST UTME ANSWERS (MATHS)

1.C 2.C 3.D 4.E 5.D 6.C 7.E 8.C 9.B 10.B 11.D 12.A 13.B 14.B 15.E 16.E 17.A 18.A
 19.A 20.C 21.D 22.B 23.E 24.C 25.C

POST-UTME

1. The factors $6x^2 - 5x$ are (a) $(x+3)$, $(x+2)$ (b) $(x-5)$, $(x-1)$ (c) $-(x+5)$, $(X+1)$ (d) $(x-5)$, $(1-x)$ (e) $(x+5)$, $(1-x)$
2. Seven years ago, the age of a father was three times that of his son, but in six years time the age of the son will be half of the father. Representing the ages of the father and son by X and Y respectively, the two equations relating X and Y are: (a) $3Y - X = 0$; $2Y - X = 0$ (b) $3Y - X = 14$; $X - 2Y = 6$ (c) $3Y - X = 7$; $X - 2Y = 6$ (d) $3Y - X = 14$; $2Y - X = 6$ (e) $X + 3Y = 7$; $X - 2Y = 12$
3. If M represents the median and D is the mode of the measurements 5, 9, 3 and 5, the (M,D) is (a) (6,5) (b) (5,8) (c) (5,7) (d) (5,5) (e) (7,5)

4. The value of $(0.03)^2 - (0.02)^2$ is (a) 0.019 (b) 0.0019 (c) 0.00019 (d) 0.000019 (e) 0.000035
5. If $263 + 441 = 714$, what number base has been used? (a) 12 (b) 11 (c) 10 (d) 9 (e) 8
6. Bala chooses at random a number between 1 and 300. What is the Probability that the number is divisible by 4 of the father and son by X and Y, respectively, the two equations relating X and Y are; a. $1/3$ b. $1/4$ c. $1/5$ d. $4/300$ e. $1/300$
7. A piece of cloth was measured as 6.10m. If the actual length of the cloth is 6.35m, find the percentage error correct to 2 decimal places. (a) 3.05% (b) 3.94% (c) 15.00% (d) 25.00% (E) 39.30%
8. Simplify $2\log_3 6 + \log_3 12 - \log 16$ (a) 2 (b) 3 (c) $22\log_3 2$ (d) $3\log_3 2^3$ (e) $4\log_3 2$
9. Find correct to two decimal places $100 + 1/100 + 3/1000 + 27/10000$ a. 100.02 b. 1000.02 c. 100.22 d. 100.01 e. 100.51
10. A bag contains 5 black, 4 white and x red marbles. If the probability of picking a red marble is $2/5$, find the value of x. a. 8 b. 10 c. 4 d. 6
11. Simplify $0.0324 \times 0.0064 / 0.48 \times 0.012$ (a) 3.6×10^2 (b) 3.6×10^{-2} (c) 3.6×10^{-3} (d) 3.6×10^{-4} (e) 3.6×10^{-5}
12. If $\cos = a/b$, find $1 + \tan a$. a. b^2 / a^2 b. A^2 / b^2 c. $(A^2 + b^2) / (b^2 + a^2)$ d. $2(a^2 + b^2) / (a^2 + b^2)$ e. B/a^2
13. If the mean of five consecutive integers is 30, find the largest of the numbers a. 28 b. 30 C. 32 d. 34
14. Simplify without using tables $(\log_2 6 - \log_2 3) / (\log_{28} 2 - \log_{21} 2)$ a. $1/5$ b. $1/2$ c. $-1/2$ d. $\log_{23} / \log_2$ e. $1/4$
15. If the 9th term of an A. P is five times the 5th term, find the relationship between a and d. a. $2a + d = 0$ b. $3a + 5d = 0$ c. $a + 3d = 0$ d. $A + 2d = 0$ e. $5d + a = 0$
16. A bucket is 12cm in diameter at the top, 8cm at the bottom and 4cm deep. Calculate its volume. a. $304/ \text{cm}^3$ b. 144 cm^3 c. $128/ \text{cm}^3$ d. 72 cm^3 e. 54 cm^3
17. Calculate $3310_5 - 1442_5$ a. 1313_5 b. 2131_5 c. 4303_5 d. 1103_5 e. 2213_5
18. Express the product of 0.0014 and 0.011 in standard form. (a) 1.54×10^{-2} (b) 1.54×10^{-3} (c) 1.54×10^{-4} (d) 1.54×10^{-5} (e) 1.54×10^{-6}
19. Factorize completely the expression $abx^2 + 6y - 3ax - 2yx$ (a) $(ax - 2y)(bx - 3)$ (b) $(bx + 3)(2y - ax)$ (c) $(bx + 3)(ax - 2y)$ (d) $(ax^2y)(ax - b)$ (e) $(ax + 1)(3 - y)$

20. Find the simple interest rate percent per annum at which N1000 accumulates to N1240 in 3 years. (A) 6% (b) 8% (c) 10% (d) 12% (e) 14%
21. An arc of a circle subtends an angle 70° at the centre. If the radius of the circle is 6cm, calculate the area of the sector subtended by the given angle. a. 22cm^3 b. 44cm^3 c. 66cm^3 d. 88cm^3 e. 108cm^3
22. A final examination requires that a student answer any 4 out of 6 questions. In how many ways can this be done? (a) 15 (b) 20 (c) 30 (d) 45 (e) 360
23. Differentiate with respect to X. $6x^2 - 5x^2 + 1/3x^2$ (a) $2 + 2/3x^2$ (b) $2 + 1/6x$ (c) $2 - 2/3x^2$ (d) $2 - 1/6x$ (e) $2 + 1/3x$
24. Divide $2x^3 + 11x^2 + 17x + 6$ by $2x$ (A) $x^2 + 5x + 6$ (b) $2x^2 + 5x + 6$ (c) $2x^3 - 5x + 6$ (d) $x^2 - 5x + 6$ (e) $x - 5x + 6$
25. If $10112 + X7 = 2510$ solve for X. (a) 14 (b) 20 (c) 24 (d) 25 (e) 28

POST UTME ANSWERS

1. D 2. C 3. D 4. D 5. D 6. D 7. B 8. C 9. D 10. A 11. D 12. B 13. A 14. C 15. E 16. C 17. D 18. A
19. D 20. A 21. B 22. A 23. E 24. C 25. A 26. A

POST UTME

26. If X is the addition of the prime numbers between 1 and 6 and Y the H.C.F of 6, 9, 15. Find the product of X and Y. (a) 27 (b) 30 (c) 33 (d) 90
27. A 5.0g of salt was weighed by Tunde as 5.1g. What is the percentage error? (a) 20 (b) 2 (c) 1 (d) 0.02
28. Two sisters, Taiwo and Kehinde, own a store. The ratio of Taiwo's share to Kehinde's is 11:9. Later, Kehinde sells $\frac{2}{3}$ of her share to Taiwo for 720.00. Find the value of the store. (a) 1,080.00 (b) 2,400.00 (c) 3,000.00 (d) 3,600.00
29. A basket contains green, black and blue balls in the ratio 5:2:1. If there are 10 blue balls, find the corresponding new ratio when 10 green and 10 black balls are removed from the basket (a) 1:1:1 (b) 4:2:1 (c) 5:1:1 (d) 4:1:1
30. A taxpayer is allowed $\frac{1}{8}$ th of his income tax free, and pays 20% on the remainder. If he pays N490.00 tax, what is his income? (A) 560.0 (b) 2,450.00 (c) 2,800.00 (d) 3,920.00

31. Evaluate $2700000 \times 0.03 + 18000$ (a) 4.5×10^0 (b) 4.5×10^1 (c) 4.5×10^2 (d) 4.5×10^3
32. The prime factors of 2,520 are (a) 2,9,5 (b) 2,9,7 (c) 2, 3,5,7 (d) 2, 3,7,9
33. If $12e = X7$, find X where $e = 12$ (a) 20 (b) 15 (c) 14 (d) 12
34. What is the difference between 0.00768,5 correct to three significant figures and 0.007685 correct to four places of decimal? (a) 10^5 (b) 7×10^{-4} (c) 8×10^{-5} (d) 10^{-5}
35. Oke deposited N800.00 in the bank at the rate of $12\frac{1}{2}\%$ simple interest. After some time the total amount was one and half times the principal. For how many years was the money left in the bank? (a) 2 (b) 4 (c) 5 $\frac{1}{3}$ (d) 8
36. If the surface area of a sphere is increased by 44%, find the percentage increase in its diameter. (a) 44 (b) 30 (c) 22 (d) 20
37. In a small village of 500 people, 350 speak the local language while 200 speak pidgin English. What percentage of the population speaks both? (A) 30% (b) 10% (c) 50% (d) 14%
38. Factorize $4a^2 - 12ab - C^2 + 9b^2$ (a) $4a(a - 3b) + (3b - c)^2$ (b) $(2a + 3b - c)(2a + 3b + c)$ (c) $(2a - 3b - c)(2a - 3b + c)$ (d) $4a(a - 3b) + (3b + c)^2$
39. What are K and L respectively if $\frac{1}{2} (3y - 4x)^2 = (8x^2 + Kxy + Ly^2)$ (a) -12, 9/2 (b) -6, 9 (c) 6, 9 (d) 12, 9/2
40. If M represents the median and D is the mode of the measurements 5, 9, 3 and 5, the (M,D) is (a) (6,5) (b) (5,8) (c) (5,7) (d) (5,5) (e) (7,5)
41. Simplify 2134×234 (a) 103 114 (b) 103214 (c) 122314 (d) 13211
42. In a class of 40 students, 32 offer Mathematics, 24 offer physics and 4 offer neither mathematics nor physics. How many offer both mathematics and physics? (a) 20 (b) 16 (c) 8 (d) 4
43. A woman buys 270 oranges for N 1,800.00 and sells at 5 for N40.00. What is her profit? (A) 2,160.00 (b) 1,620.00 (c) 630.00 (d) 360.00
44. Find the simple interest rate percent per annum at which N1000 accumulates to N1240 in 3 years. (a) 6% (b) 8% (c) 10% (d) 12% (e) 14%
45. The length a person can jump is inversely proportional to his weight. If a 20kg person can jump 1.5m, find the constant of proportionality. (a) 60 (b) 30 (c) 20 (d) 15
46. Find the values of X and Y respectively if $3x - 5y + 5 = 0$ and $4x - 7y + 8 = 0$ (a) 0,4 (b) -4,-5 (c) 4,5 (d) 0.5,4
47. Divide $6x^2 - 13x + 5$ by $2x - 1$ (a) $3x + 5$ (b) $3x - 5$ (c) $5x - 3$ (d) $5x + 3$

48. A polynomial in X whose zeros are $-2, -1$ and 3 is (A) $x^3 - 7x + 6$ (b) $x^3 + 7x - 6$ (c) $x^3 + 7x + 6$ (d) $x^3 - 7x - 6$

49. The time taken to do a piece of work is inversely proportional to the number of men employed. If it takes 30 men to do a piece of work in 6 days, how many men are required to do the work in 4 days? (A) 20 (b) 35 (c) 5 (d) 60

POST UTME ANSWERS

25.B 26.B 27.B 28.A 29.B 30.D 31.B 32.A 33.C 34.C 35.D 36.A 37.B 38.C 39.C
40.D 41.D 42.C 43.A 44.B 45.A 46.B 47.C 48. 49. D 50. C

POST UTME

POST UTME

1. A student measures a piece of rope and found that it was 1.26 m long. If the actual length of the rope was 1.25 m, what was the percentage error of the measurement? A. 0.40% B. 0.01% C. 0.25% D. 0.80%.

2. At what rate will the interest on N400 increase to N 24 in 3 years reckoning in simple interest? A. 4% B. 2% C. 3% D. 5%.

3. Given that $\log 2 = 0.3010$, $\log 7 = 0.8451$. Evaluate $\log 112$. A. 2.5441 B. 2.0491 C. 2.1461 D. 3.1461.

4. Express the product of 0.21 and 0.34 in standard form A. 7.14×10^{-3} B. 7.14×10^{-1} C. 7.14×10^{-2} D. 7.14×10^{-4}

5. In a survey of 50 newspaper readers, 40 read Champion and 30 read Guardian, how many read both papers? A. 15 B. 5 C. 10 D. 20.

6. If $9x^2 + 6xy + 4y^2$ is a factor of $27x^3 - 8y^3$, find the other factor. A. $3x - 2y$ B. $2y - 3x$ C. $2y + 3x$ D. $3x + 2y$

7. Solve for x and y if $x - y = 2$ and $x^2 - y^2 = 8$ A. (1, 3) B. (3, 1) C. (-1, 3) D. (-3, 1).

8. If y varies directly as the square root of x and $y = 3$ when $x = 16$. Calculate y when $x = 64$. A. 5 B. 12 C. 6 D. 3

9. If x is inversely proportional to y and $x = 2\frac{1}{4}$ when $y = 2$, find x if $y = 4$. A. $2\frac{1}{4}$ B. 5 C. 4 D. $1\frac{1}{4}$
10. Find the sum to infinity of the following series. $0.5 + 0.05 + 0.005 + 0.0005\ldots$
A. $\frac{5}{9}$ B. $\frac{5}{7}$ C. $\frac{5}{11}$ D. $\frac{5}{11}$
11. The 3rd term of an arithmetic progression is -9 and the 7th term is -29. Find the 10th term of the progression. A. 44 B. -165 C. -44 D. 165.
12. If the length of a square is increased by 20% while its width is decreased by 20% to form a rectangle, what is the ratio of the area of the rectangle to the area of the square? A. 6:5 B. 25:24 C. 5:6 D. 24:25
13. Two brothers invested a total sum of N5, 000.00 on a farm project. The farm yield was sold for N15,000.00 at the end of the season. If the profit was shared in the ratio 2:3, what is the difference in the amount of profit received by the brothers? A. N2, 000.00 B. N4, 000.00 C. N6, 000.00 D. N10,000.00.
14. Peter's weekly wages are N20.00 for the first 20 weeks and N36.00 for the next 24 weeks. Find his average weekly wage for the remaining 8 weeks of the year, if his average weekly wage for the whole year is N30.00. A. N37.00 B. N35.00 C. N30.00 D. N25.00
15. A man invests a sum of money at 4% per annum simple interest. After 3 years the principal amounts to N7, 000.00. Find the sum invested. A. N7, 840.00 B. N6, 250.00 C. N6,160.00 D. N5,833.33.
16. By selling 20 oranges for N1.35 a trader makes a profit of 8%. What is his percentage gain or loss if he sells the same 20 oranges for N1.10? A. 8% B. 10% C. 12% D. 15%.
17. Four boys and ten girls can cut a field in 5 hours. If the boys work at $\frac{5}{4}$ hrs rate of which the girls work, how many boys will be needed to cut the field in 3 hours? A. 180 B. 60 C. 25 D. 20
18. The factors $6x^2 - 5x - x^2$ are (a) $(x+3), (x+2)$ (b) $(x-5), (x-1)$ (c) $-(x+5), (X+1)$ (d) $(x-5), (1-x)$ (e) $(x+5), (1-x)$
19. Seven years ago, the age of a father was three times that of his son, but in six years time the age of the son will be half of the father. Representing the ages of the father and son by X and Y , respectively, the two equations relating X and Y are; (a) $3Y - X = 0; 2Y - X = 0$ (b) $3Y - X = 14; X - 2Y = 6$ (c) $3Y - X = 7; X - 2Y = 6$ (d) $3Y - X = 14; Y - 2X = 6$ (e) $X + 3Y = 7; X = 2Y = 12$
20. If M represents the median and D is the mode of the measurements 5, 9, 3 and 5, the (M,D) is (a) (6,5) (b) (5,8) (c) (5,7) (d) (5,5) (e) (7,5)

21. The value of $(0.03)^2 - (0.02)^2$ is (a) 0.019 (b) 0.0019 (c) 0.00019 (d) 0.000019 (e) 0.000035

22. If $263 + 441 = 714$, what number base has been used? (a) 12 (b) 11 (c) 10 (d) 9 (e) 8

23. What is the probability of having an odd number in a single toss of a fair die? A. $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$ (e) $\frac{5}{6}$

24. A sales girl gave a change of N1.15 to a customer instead of N1.25. Calculate the Percentage error (a) 10% (b) 8.7% (c) 8.0% (d) 2.4% (e) 0.10%

25.

Score	0	1	2	3	4	5
Frequency	2	3	4	2	7	2

Find the mean score in the frequency table above showing the performance of a group of students in a test. (A) 1.75 (b) 2 (c) 2.5 (d) 2.75 (e) 3

POST UTME ANSWERS

51. D 52. B 53. B 54. C 55. D 56. A 57. B 58. C 59. D 60. A 61. C 62. 63. A 64. A 65. B 66. C 67. D 68. C 69. D 70. D 71. D 72. D 73. C 74. C 75. D

MATHEMATICS

- 1 A regular polygon have each of its number of sides of the polygon?
(a) 36 (b) 9 (c) 18 (d) 20
- 2 A girl walks 30m from a point P on a bearing of 0400 to a point Q. She the walks 30m on a bearing of 1400 to a point R. The bearing of R from P is
(a) 900 (b)500 (c) 450 (d) 400
- 3 How many different three digit numbers can be formed using the integers 1 to 6 if no integer occurs twice in a number? (a) 245 (b)120 (c) 60 (d) 48
- 4 In how many different ways can the letters of the word geology be arranged in
(a)720 (b)1260 (c)25200 (d)5040
- 5 A cyclist rode for 30 minutes at xhm/h and due to a breakdown he has to push the byke for 2hrs at $x - 5$ km.hr. If the total distance covered is less than 60km, what is the range of values for x? (a) $x < 14$ (b) $x < 20$ (c) $x < 29$ (d) $x < 28$
- 6 A businessman invested a total of N200, 000 in two companies which paid dividends of 5% and 7% respectively. If he received a total of N11,600 as dividend, how much did he invested at 5%? (a) N160,000 (b) N140,000 (c) N120,000 (d) N80,000
- 7 In a class, 37 students take at least one of chemistry, Economics and Government, 8 students take Chemistry, 19 take Economics and 25 take Government. 12 students take Economics and Government but nobody takes Chemistry and Economics. How many students take both Chemistry and Government? (a) 3 (b) 4 (c) 5 (d) 6
- 8 Solve for p in the following equation given in base two: $11(p + 110) = 1001p$ (a) 10 (b) 11 (c) 110 (d) 111
- 9 Factorize $16(3x + 2y)^2 - 25(a + 2b)^2$. $(12x + 8y + 5a + (a) 10b) (12x + 8y - 5a - 10b)$
 $(12x + 8y - 5a (b) 10b) (12x + 8y - 5a - 10b (c) 20 (3x + 2y - a - 2b) (3x + 2y + a + 2b)$
(d) $20 (3x + 2y + a + 2b) (3x + 2y + a + 2b)$
- 10 A cone has base radius 4cm and height 3cm. the area of its curved surface is (a) $12\pi\text{cm}^2$ (b) $24\pi\text{cm}^2$
(c) $20\pi\text{cm}^2$ (d) $15\pi\text{cm}^2$
- 11 Let $\log y + 3 \log x = 3$, then, y is (a) $(10/x)^2$ (b) $(x/10)^3$ (c) $(x/10)^{-3}$ (d) $(10/x)^{-1/3}$
- 12 If $x^2 - y^2 = 1$, then y is $a^2 b^2$ (a) $\pm a x^2 - a^2$ (b) $a a^2 - x^2 b b$ (c) $\pm a x^2 - a^2$ (d) $b a^2 - x^2 b a$
- 13 z is partly constant and partly varies inversely as the square of d. When $d = 1$, $z = 11$ and when $d = 2$, $z = 5$. Find the value of z when $d = 4$ (a) 2 (b) 3.5 (c) 5 (d) 5.5
- 14 Expand the expression $(x^2 - 2x - 3)(x^2 + x + 1)$
(a) $x^4 - 4x^2 - 5x - 3$ (b) $-x^3 - 4x^2 + 5x - 3$ (c) $X^4 - x^3 - 4x^2 - 5x - 3$ (d) $X^4 - 4x^2 - 5x - 2$
- Suppose we have matrices $A = \begin{pmatrix} 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \end{pmatrix}$
- 15 find $A^2 + AB - 2A$

$$\begin{pmatrix} 2 & 3 & 4 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 & 8 & 7 \end{pmatrix}$$

$$\begin{pmatrix} 12 & 13 & -8 & -6 \end{pmatrix}$$

(a) 5 - 9 (b) -1 -4 (c) - 4 - 4 (d) 0 -4

16 The inverse of matrix B is (a) $\begin{pmatrix} 1 & -3 & 2 \\ -3 & 2 & 8 \\ 4 & 0 & 4 \end{pmatrix}$ (b) $\begin{pmatrix} -3 & 2 & 8 \\ 4 & 0 & 4 \\ 1 & 3 & -4 \end{pmatrix}$ (c) $\begin{pmatrix} 1 & 3 & -4 \\ 1 & 0 & 8 \\ -3 & 2 & 4 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 0 & 8 \\ -3 & 2 & 4 \\ 1 & 3 & -4 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 3 & -4 \\ 1 & 0 & 8 \\ -3 & 2 & 4 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 0 & 8 \\ -3 & 2 & 4 \\ 1 & 3 & -4 \end{pmatrix}$

(12x + 8y + 5a +

(a) 10b) (12x + 8y - 5a - 10b)

(12x + 8y - 5a (b) 10b) (12x + 8y - 5a - 10b

(c) 20 (3x + 2y - a - 2b) (3x + 2y + a + 2b)

(d) 20 (3x + 2y + a + 2b) (3x + 2y + a + 2b)

10 A cone has base radius 4cm and height 3cm. the area of its curved surface is (a) $12\pi\text{cm}^2$ (b) $24\pi\text{cm}^2$ (c) $20\pi\text{cm}^2$ (d) $15\pi\text{cm}^2$

11 Let $\log y + 3 \log x = 3$, then, y is (a) $(10/x)^2$ (b) $(x/10)^3$ (c) $(x/10)^{-3}$ (d) $(10/x)^{-1/3}$

12 If $x^2 - y^2 = 1$, then y is $a^2 b^2$

(a) $\pm a x^2 - a^2$ (b) $a a^2 - x^2 b b$

(c) $\pm a x^2 - a^2$ (d) $b a^2 - x^2 b a$

13 z is partly constant and partly varies inversely as the square of d. When $d = 1$, $z = 11$ and when $d = 2$, $z = 5$. Find the value of z when $d = 4$ (a) 2 (b) 3.5 (c) 5 (d) 5.5

14 Expand the expression $(x^2 - 2x - 3)(x^2 + x + 1)$

(a) $x^4 - 4x^2 - 5x - 3$

(b) $-x^3 - 4x^2 + 5x - 3$

(c) $x^4 - x^3 - 4x^2 - 5x - 3$

(d) $x^4 - 4x^2 - 5x - 2$

Suppose we have matrices

$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix}$

$\begin{pmatrix} 2 & 3 \\ 4 & 3 \end{pmatrix}$

15 find $A^2 + AB - 2A$

(a) 5 - 9 (b) -1 -4

$\begin{pmatrix} 2 & 3 \\ 8 & 7 \end{pmatrix}$

(c) - 4 - 4 (d) 0 -4

$\begin{pmatrix} 12 & 13 \\ -8 & -6 \end{pmatrix}$

$\begin{pmatrix} 13 & 3 \\ 3 & 1 \end{pmatrix}$

16 The inverse of matrix B is (a) $\begin{pmatrix} 1 & -3 & 2 \\ -3 & 2 & 8 \\ 4 & 0 & 4 \end{pmatrix}$ (b) $\begin{pmatrix} -3 & 2 & 8 \\ 4 & 0 & 4 \\ 1 & 3 & -4 \end{pmatrix}$ (c) $\begin{pmatrix} 1 & 3 & -4 \\ 1 & 0 & 8 \\ -3 & 2 & 4 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 0 & 8 \\ -3 & 2 & 4 \\ 1 & 3 & -4 \end{pmatrix}$

$\begin{pmatrix} 8 & 4 & 0 & 4 & 0 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 3 & -4 \\ 1 & 0 & 8 \\ -3 & 2 & 4 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 0 & 8 \\ -3 & 2 & 4 \\ 1 & 3 & -4 \end{pmatrix}$

$\begin{pmatrix} 8 & -2 & 0 & 0 & 1 \end{pmatrix}$

17 Find indefinite integral of the function $f(x) = x \cos x$ for any constant k, is

(a) $x \cos x + \sin x + k$

(b) $x \sin x - \cos x$

(c) $x \sin x + \cos x + k$

(d) $x \sin x - \cos x + k$

18 Evaluate the integral $\int_2^3 (x^2 + 1/x) dx$

(a) $8 + \ln 2$ (b) $7 + \ln 2$

3 3

(c) $7 - \ln 3$ (d) 8

3 3

19 The trigonometric expression $\cos 2A + \sin 2A$ can be written as

(a) $\cos A (\cos A - \sin A)$

(b) $\cos^2 A + \sin^2 A - 2 \sin A \cos A$

(c) $2 \sin A \cos A + \cos^2 A + \sin^2 A$

(d) $\cos^2 A + \sin^2 A + \cos A$

Suppose D, E and P are subsets of a universal set U. U be the set of natural numbers no greater than 10, while D, E and P are respectively the set of odd numbers, even numbers and prime numbers. For any set x, its complement $\bar{x}$ and $\emptyset$ denote the empty set.

20 Display the set $D \cap P$. (a) {3,5,7} (b) {2} (c) {1,2,3,4,5,6,7,8,9,10} (d) $\emptyset$

A bag contains 10 balls of which 3 are red and 7 are white. Two balls are drawn at random. Find the probability of none of the balls is red if the draw is.

22 With replacement: (a) 0.9 (b) 1 (c) 0.4 (d) 0.49

23 Without replacement: (a) 0.1 (b) 0.47 (c) 0.42 (d) 0.21

ANSWERS

1D 2A 3B 4B 5 D 6C 7C 8B 9A 10C 11A 12A 13B 14C 15A 16A 17A 18A 19C 20A

21D 22D 23D

MATHEMATICS

1. Evaluate $2 \log_3 + 3 \log_2 - 2 \log_6$

A. $2 \log_3$ B.

2. If $31410 - 2567340x$ find X A.9 B. 10 C. 7 D. 8

3. Determine $2.813 \times 10^3 \times 1.063/5.637 \times 10^{-2}$ reducing each number to 2s.f and leaving the answer in 2s.f. A.0.54 B.0.054 C. 0.0056 D.0.055

4. Solve the inequality $2-x > x^2$. A. $x-2a < 1$ B. $x > 2$ or $x < -1$ C. $x < -2$ or $x > 1$ D. In the diagram f(x) is represented by $(3xR)0(2y7)0$

5. In the diagram f(x) is represented by: A. x^2-2x+3 B. $x^2-i-3x+2$ C. $x+3x+2$ D. x^2-2x+3

6. Find the minimum value of the function $f(\theta) = 2 - 3 \cos \theta$, for $0 \leq \theta \leq \pi$

A. $1/2$ B. $2/3$ C.2 D. 1A. $513/1024$ B. $615/1024$ C. $968/1024$ D. $1011/1024$

8. A man bought 220 mangoes at #5x. He sold each for 3x kobo and made a gain of #8. Find the value of x. A.2 B. 5 C.10 D. 6

9. Calculate the sum to infinity of $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$ A. 0.33 B. 0.67 C. 1.5 D. 2.5

10. If $P + C^2 = 9$, then n can be.

A. 1 B. 2 C. $3/2$ D. 3

11. Express $\tan 22\frac{1}{2}^\circ$ in the form $A + \sqrt{2} - 2$ B. $1 - \sqrt{2}$ C. $\sqrt{2} - 1$ D. $\sqrt{2} + 1$
12. If $\sin(x-a) = \cos(x+a)$, then $\tan a$ is: A. 0.8 B. 0.75 C. 1.0 D. 2.8
13. Simplify 110113_A , 101000_a B. 11003 C. 1110 , D. 10112
14. A binary operation x is defined on R , the set of real numbers by $xy = vxy$, for all $x, y \in R$. If $x*(2*8)=6$, find, A. 2 B. 4 C. 9 D. 10
15. The equation $2x^2 + (k+3)x + 2k = 0$ has co-incident roots. What is the constant k ? A. 3 or 7 B. 4 or 0 C. 9 or 1 D. 9 or 1
16. The curve $y = 2x^2 + x - 2$ and the straight line $y = 2x - 1$ meet at the points whose x values are: A. $1/2$ and 1 B. 2 and . C. $1/2$ and 1 D. 2, .
17. Find the mean deviation of 5, 8, 3, 7 and 2. A. 0 B. 2 C. 5 D. 10
18. The sequence 'a, -1, -1/256' has n terms. Determine the value of n . A. 6 B. 7 C. 8 D. 9
19. The volume of two similar solid cubes are 729cm^3 and 512cm^3 . Find the ratio of their lengths. A. 4:3 B. 9:7 C. 8:7 D. 9:8
20. In fig C, PR is a diameter, $\angle PRQ = (3x-8)^\circ$ and $\angle RPQ = (2y-7)^\circ$ Express x in terms of y . A. $x = 75 + 2y$ B. $x = 105 + 3y$ C. $x = 105 - 2y$ D. $x = 75 - 3y$
21. Find the mean deviation of 6, 7, 8, 9, 10 A. 1.2 B. 1.5 C. 2 D. 8
22. The matrix is $2I$, $34I$ If I is the 2×2 unit matrix, then $k^2 + 1 + k$ is: A. 6 B. 3 C. 13 D. 20 B. 9 C. 8 D. 22 C. 10 D. 7 E. 21 F. 24 D. 1 G. 4 H. 25
23. The line joining $(p, 4)$ and $(6, -2)$ is perpendicular to the line joining $(2, p)$ and $(-1, 3)$. Find p . A. 6 B. 3 C. 4 D. 0
24. In a regular polygon, each interior angle double its corresponding exterior angle. Find the number of sides of the polygon. A. 3 B. 4 C. 6 D. 8
25. The mean of the numbers 0, $x+2$, $3x+6$ and $4x+8$ is 4. Find their mean deviation, A. 4 B. 3 C. 2 D. 0

MATHEMATICS

1D 2C 3D 4A 5A 6A 7C 8B 9C 10B 11C 12C 13C 14C 15C 16A 17B 18C 19D 20C 21B 22C 23C 24C 25B

MATHEMATICS

1. In a bag, there are 8 identical balls in terms of size and shape, but 2 are white, 4 are black and 2 are yellow. If balls are randomly drawn from the bag without replacement, what is the probability that two balls drawn from the bag, one is white and the other black? A. $2/7$ B. $6/7$ C. $5/7$ D. $1/7$
2. If X_0 is acute and $\cos 2x = 0$ A. 144
3. Evaluate 1 without using tables $811/2$ A. 234 B. -729 C. 243
4. Find the roots of the equation $6x^2 - 13x + 6 = 0$ A. $2/3, 3/2$ B. $2/3, -3/2$ C. $-2/3, -3/2$ D. $2/3, 3/2$
5. A polynomial in x whose zeros are 2, -1 and -7 is $x^2 + 6x^2 + 9x + 14bx^3 - 6x^2 + 14cx^3 + 6x^2 - 9x + 14dx^3 - 6x^2 - 9x + 14$

6. If $y = 91 - 3x^3$, find the value of dy at dx^2 . $x = -2$ A. 378 B. -441 C. -378 D. 441
6. If $y = (1 - 3x)^3$, find the value of dy at dx^2 $x = -2$ A. 378 B. -441 C. -378 D. 441
7. Convert 366 base ten to a binary number A. 101011001 B. -101101110 C. 10110111
8. Simplify $8x^5 \cdot 2x^2 - 25x - 6$ A. $4x - 1$ B. $4x - 12x$ C. $2x + 3(2x - 3)$
9. Given that $x = 3$, $y = -2$ and $z = -1$. Evaluate $3x^2y^2zx^2y^3z$ A. 6 B. 9 C. -6 D. -9
10. In how many ways can 9 rubber seats be arranged if three are of the same colour and size? A. 48 B. 720 C. 420 D. 84
11. Two ships A and B left port P on a bearings 0530 and 1130 respectively. If both ships steamed at 80km per hour, how far apart were they after 30 minutes? A. -80km B. 60km C. 20km D. 40km
12. What are the coordinated of the midpoint of the line joining (-1,6) and (5,-3) A. (3,4,5) B. (2,1 .) C. (-3,4,5) D. (-2,-1.5)

MATHEMATICS

1A 2B 3B 4C 5A 6B 7A 8C 9C 10 11C 12D

MATHEMATIC FOUR

- 1 the interior angles of a pentagon (a) 75° (b) 108° (c) 120° (d) 134°
2. All the vertices of an isosceles triangle lie on a circle and each of the base angles of the triangle is 65° . The angle subtended at the centre of the circle by the base of the triangle is: (a) 130° (b) 115° (c) 100° (d) 65°
3. A square tile measures 20cm by 20cm. How many of such tiles will cover a floor measuring 5m by 4m? (a) 500 (b) 400 (c) 320 (d) 250
4. The volume of a certain sphere is numerically equal to twice its surface area. The diameter of the sphere is: (a) 6 (b) 9 (c) 12 (d) $\sqrt{v}$
5. A bearing of 310° , expressed as a compass bearing is: (a) $N50^\circ W$ (b) $N40^\circ W$ (c) $S40^\circ W$ (d) $S50^\circ W$
6. Which of the following specified sets of data is not necessarily sufficient for the construction of a triangle? (a) three angles (b) two sides and a right angle (c) two sides and an included angle (d) three sides
7. The average age of the three children in a family is 9 years. If the average age of their parent is 39 years, the average age of the whole family is: (a) 20 years (b) 21 years (c) 24 years (d) 27 years
10. Evaluate x in base 3 if $41_{-}22_{-}=17$ (a) 11 (b) 8 (c) 12 (d) 22
11. A woman buys 4 bags of rice for ₦56 per bag and 3 bags of beans for ₦26 per bag using the currency "LONI" (b) in base 7. What is the total cost of the items in another currency "MONI" (M) in base 8? (a) M224 (b) M114 (c) N340 (d) M440
12. When the price of egg was raised by ₦2 an egg, the number of eggs which can be bought for ₦120 is reduced by 5. The present price of an egg is

(a) N6 (b) N7 (c) N8 (d) N10

13. How long will it take a sum of money invested at 8% simple interest to double the (d) 134°

2. All the vertices of an isosceles triangle lie on a circle and each of the base angles of the triangle is 65°.

The angle

subtended at the centre of the circle by the base of the triangle is:

(a) 130° (b) 115° (c) 100°

(d) 65°

3. A square tile measures 20cm by

20cm. How many of such tiles will cover a floor measuring 5m by 4m? (a) 500

(b) 400

(c) 320 (d) 250

4. The volume of a certain sphere is numerically equal to twice its surface area. The

diameter of the sphere is:

(a) 6 (b) 9 (c) 12 (d) 15

5. A bearing of 310°, expressed as a compass bearing is:

(a) N50°W (b) N40°W

(c) S40°W (d) S50°W

6. Which of the following specified sets of data is not necessarily sufficient for the

construction of a triangle?

(a) three angles

(b) two sides and a right angle (c) two sides and an included angle (d) three sides

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7. The average age of the three children in a family is 9 years. If the

average age of their

parent is 39 years, the average age of the whole family is:

(a) 20 years (b) 21 years

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10. Evaluate x in base 3 if

$41_{(3)} - 22_{(3)} = 17_{(3)}$

(a) 11 (b) 8 (c) 12 (d) 22

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using the currency "LONI" (b) in base 7. What is the total cost of the items in another

currency "MONI" (M) in base 8?

(a) M224 (b) M1 14 (c) N340 (d) M440

12. When the price of egg was raised by N2 an egg, the number of eggs which can be bought for N120 is reduced by 5. The present price of an egg is

(a) N6 (b) N7 (c) N8 (d) N10

13. How long will it take a sum of money invested at 8% simple interest to double the original sum?

(a) 8 years (b) 10.5 years (c) 12 years (d) 12.5 years

14. The journey from Lagos to Ibadan usually takes motorist 1 hour 3 minutes.

By

increasing his average speed by 20km/hr, the motorist saves 15 minutes. His usual speed, in km/hr is

(a) 100 (b) 90 (c) 85 (d) 80

15. The smallest section of a rod which can be cut into exactly equal sections, each of either 30cm or 36cm in length is

(a) 90cm (b) 180cm (c) 360cm (d) 540cm

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16. If $x = 0.0012 \text{ --- } 0.00074\text{-i- } 0.003174$, what is the difference between x to 2 decimal

places and x to 1 significant figure?

(a) 0.01 (b) 0.005 1 (c) 0.1 (d) 0.005

17. The angle of depression of two points A and B on a plane field from the top of a mast erected between A and B are 30° and 45° respectively. If A is westward of B, find IABI if the height of the mast is 15m from the field.

18. The radius of a circle is given as 10cm subject to an error of 0.2cm. The error in the area of the circle is (a) -% (b) -% (c) 2% (d) 4%

19. If θ is acute, evaluate (a) $\tan \theta$ (b) $-\tan \theta$ (c) $\cot \theta$ (d) $-\cot \theta$

20. In a survey of 100 students in an institution, 80 students speak Yoruba,

22 speak Igbo, while 6 speak neither Igbo nor Yoruba. How many students speak Yoruba and Igbo? (a) 96 (b) 8 (c) 64 (d) 12

21. A bag contains 5 yellow balls, 6 green balls and 9 black balls. A

ball is drawn from the bag. What is the probability that it is a black or

yellow ball? The table below shows the distribution of weight measure for 100 students. Weight (kg

60-

62

63-

65

(d) 134°

2. All the vertices of an isosceles triangle lie on a circle and each of the base angles of the triangle is 65° .

The angle subtended at the centre of the circle by the base of the triangle is:

(a) 130° (b) 115° (c) 100° (d) 65°

3. A square tile measures 20cm by 20cm. How many of such tiles will cover a floor measuring 5m by 4m? (a) 500 (b) 400 (c) 320 (d) 250

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(a) $N50^\circ W$ (b) $N40^\circ W$ (c) $S40^\circ W$ (d) $S50^\circ W$

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7. The average age of the three children in a family is 9 years. If the average age of their

parent is 39 years, the average age of the whole family is:

(a) 20 years (b) 21 years (c) 24 years (d) 27 years

10. Evaluate x in base 3 if $41_{(3)} - 22_{(3)} = 17_{(3)}$ (a) 11 (b) 8 (c) 12 (d) 22

11. A woman buys 4 bags of rice for ₦56 per bag and 3 bags of beans for ₦26 per bag using the currency "LONI" (L) in base 7. What is the total cost of the items in another currency "MONI" (M) in base 8? (a) $M224$ (b) $M114$ (c) $N340$ (d) $M440$

12. When the price of egg was raised by ₦2 an egg, the number of eggs which can be bought for ₦120 is reduced by 5. The present price of an egg is

(a) ₦6 (b) ₦7 (c) ₦8 (d) ₦10

13. How long will it take a sum of money invested at 8% simple interest to double the original sum? (a) 8 years (b) 10.5 years (c) 12 years (d) 12.5 years

14. The journey from Lagos to Ibadan usually takes motorist 1 hour 3 minutes. By increasing his average speed by 20km/hr, the motorist saves 15 minutes. His usual speed, in km/hr is (a) 100 (b) 90 (c) 85 (d) 80

15. The smallest section of a rod which can be cut into exactly equal sections, each of either 30cm or 36cm in length is (a) 90cm (b) 180cm (c) 360cm (d) 540cm

16. If $x = 0.0012 - 0.00074i - 0.003174$, what is the difference between x to 2 decimal places and x to 1 significant figure? (a) 0.01 (b) 0.0051 (c) 0.1 (d) 0.005

17. The angle of depression of two points A and B on a plane field from the top of a mast erected between A and B are 30° and 45° respectively. If A is westward of B, find IABI if the height of the mast is 15m from the field.

18. The radius of a circle is given as 10cm subject to an error of 0.2cm. The error in the area of the circle is (a) -% (b) +% (c) 2% (d) 4%

19. If θ is acute, evaluate (a) $\tan \theta$ (b) $-\tan \theta$ (c) $\cot \theta$ (d) $-\cot \theta$
20. In a survey of 100 students in an institution, 80 students speak Yoruba, 22 speak Igbo, while 6 speak neither Igbo nor Yoruba. How many students speak Yoruba and Igbo? (a) 96 (b) 8 (c) 64 (d) 12
21. A bag contains 5 yellow balls, 6 green balls and 9 black balls. A ball is drawn from the bag. What is the probability that it is a black or yellow ball?
- The table below shows the distribution of weight measure for 100 students.

Weight

(kg

60-

62

63-

65

66-

68

69-

71

71-

74

f 5 18 42 27 8

22. Calculate the mean of the distribution to two decimal places.

(a) 64.45 (b) 62.45 (c) 67.45 (d) 65.45

23. Calculate the mode of the distribution to two decimal places

(a) 67.33 (b) 65.33 (c) 65.53 (d) 67.35

ANSWER

1D 2C 3A 4C 5A 6A 7B 8A 9C 10B 11C 12C 13D 14A 15B 16D 17C 18D 19B 20B

21D 22C 23A